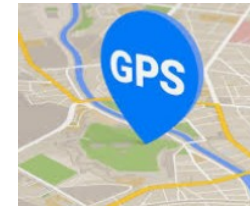
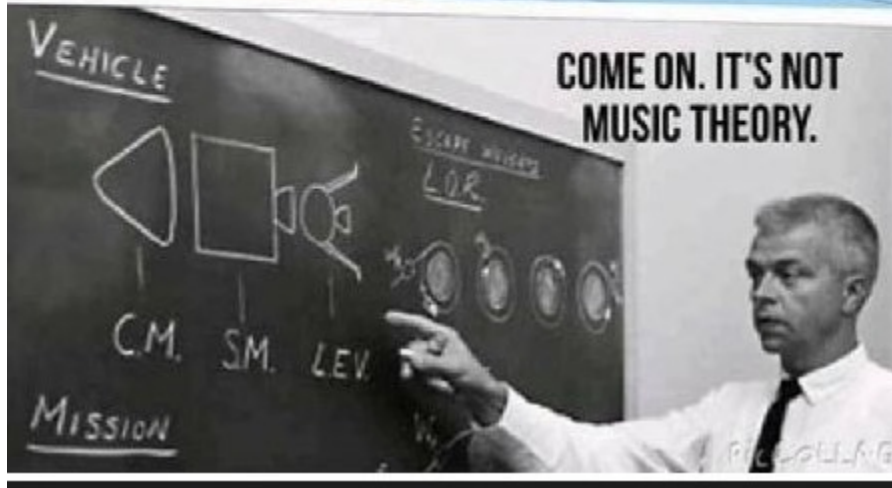


Why you need to take Physics → Motivational speech!

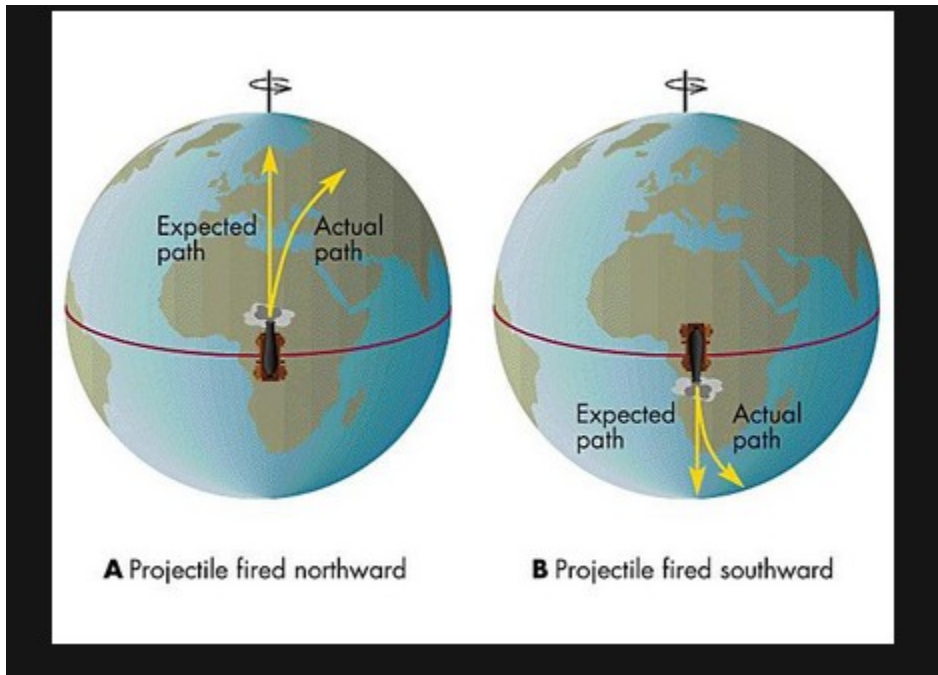
Physics lays down the foundations of technology

Frame of
Reference!



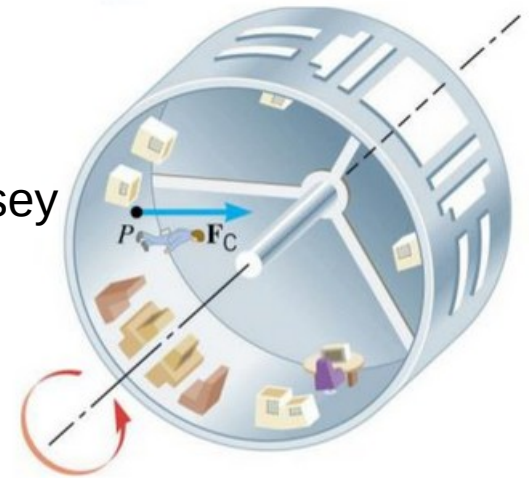
Little Boy and Fat Man | P...

Rotating frames of reference → Coriolis force and centrifugal force
(also called pseudo forces. Happen in a non inertial frame of reference)

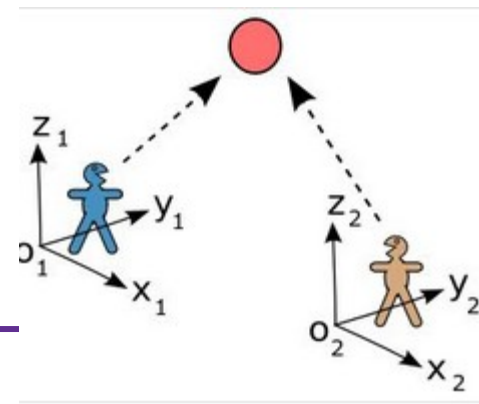


centripetal force or centrifugal force?
A matter of frame of reference.

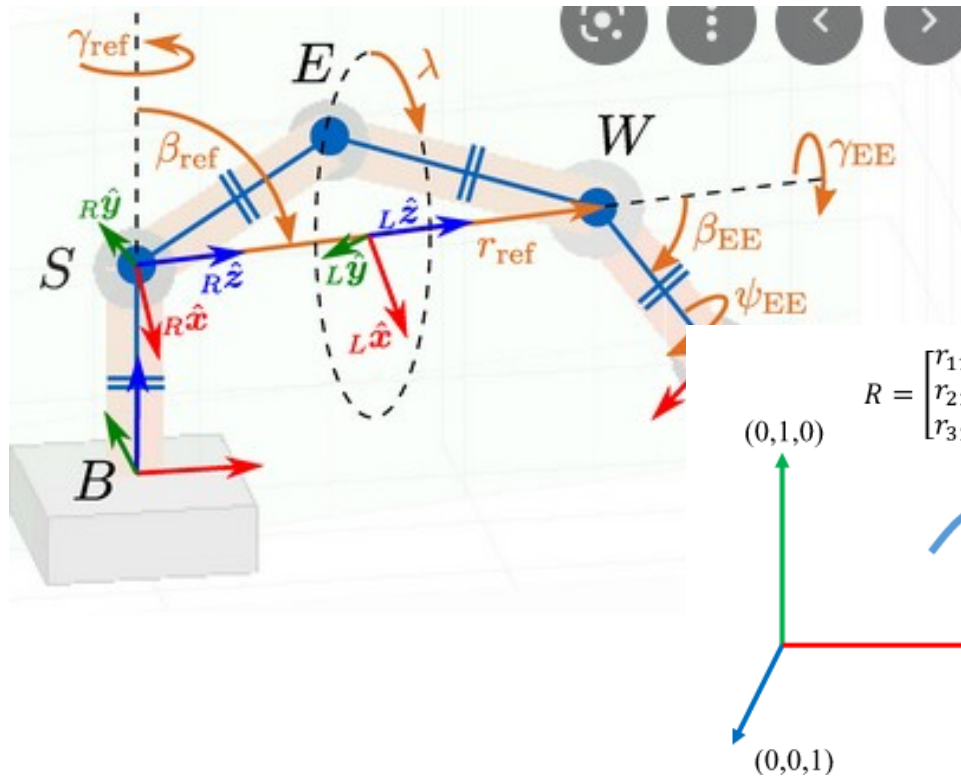
See
Space odyssey
movie



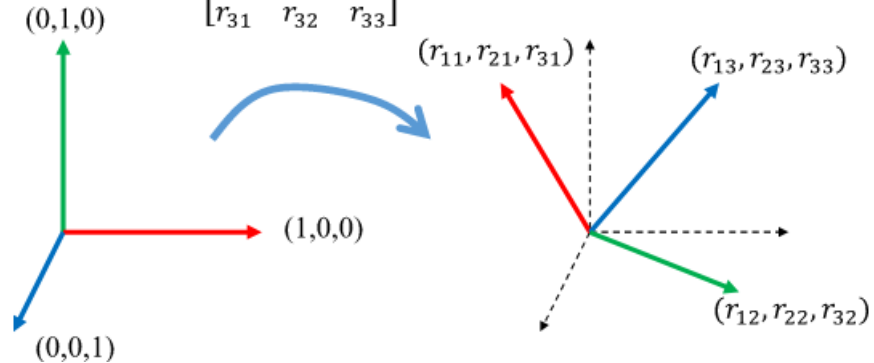
See video. In Physics, the frame of reference matters!



Frames of reference in coding and robotics.



$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

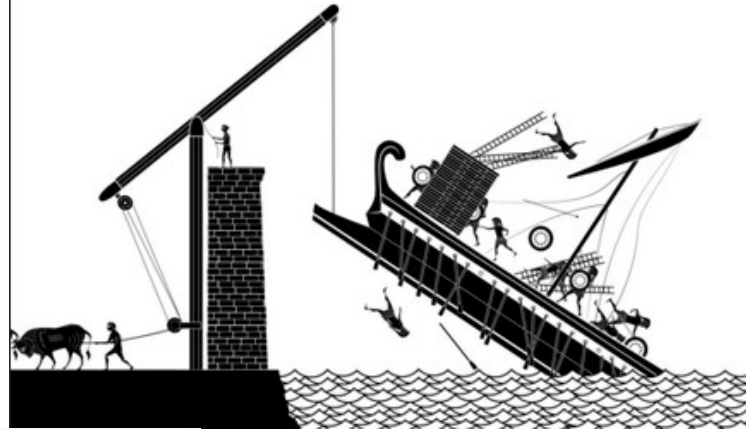
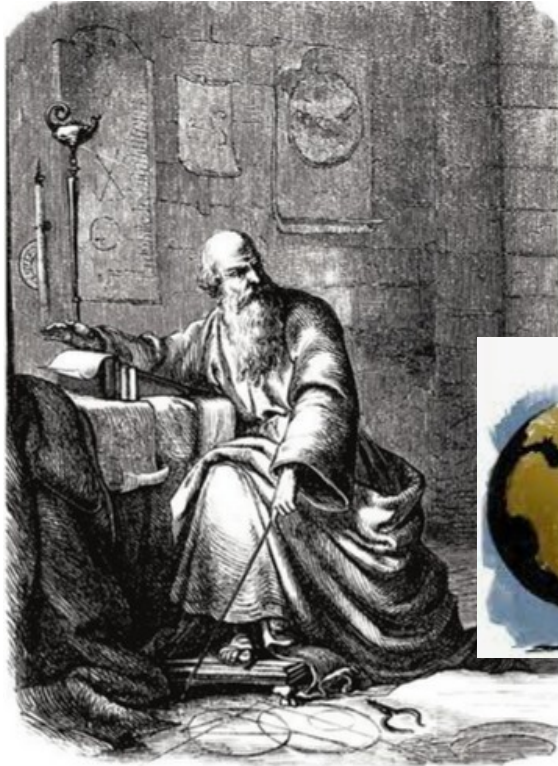


Galileo (1564-1642) the laws of nature are written in the language of mathematics. This was also noted thousands of years earlier by Greeks, Babylonians and Indians

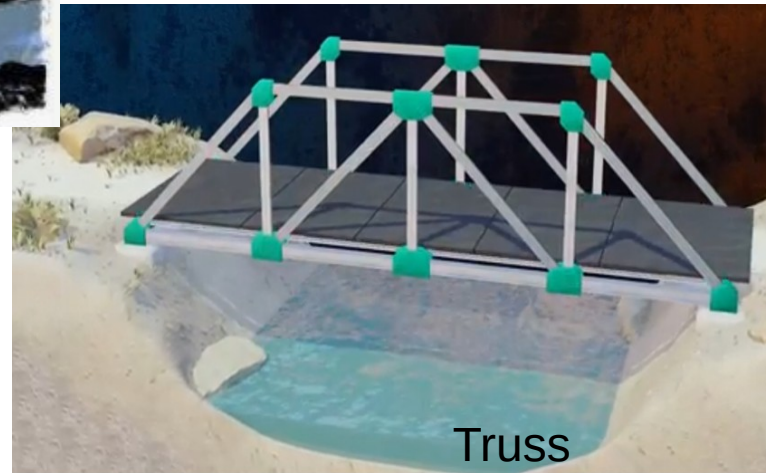
Bing a TI 83 (TI 30) in class!
Review Sci Notation, proportions
And trig.

Mechanics – Equilibrium (static) – Machines (conservation of energy)

And torques



Give me a lever long enough
and a fulcrum on which to place it,
and I shall move the world.



Truss
bridge

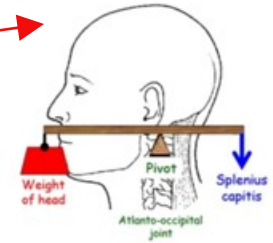
Type of Levers in Human Body



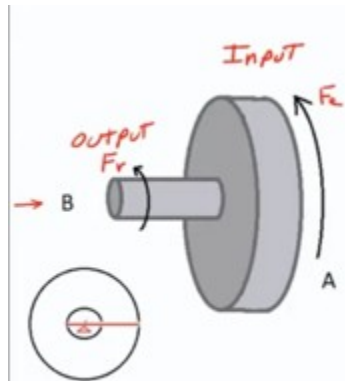
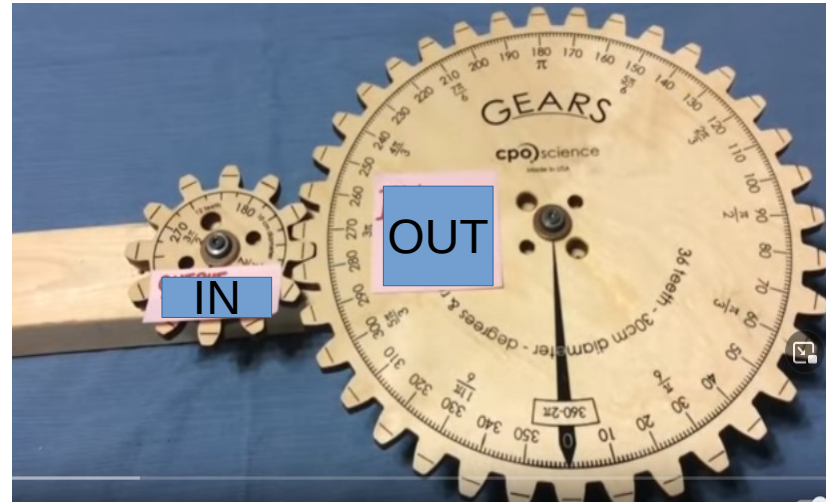
- First Class Lever
 - Example: Head-Neck Jo

- Second Class Lever
 - Example: Ankle Joint

- Third Class Lever
 - Example: Elbow Joint

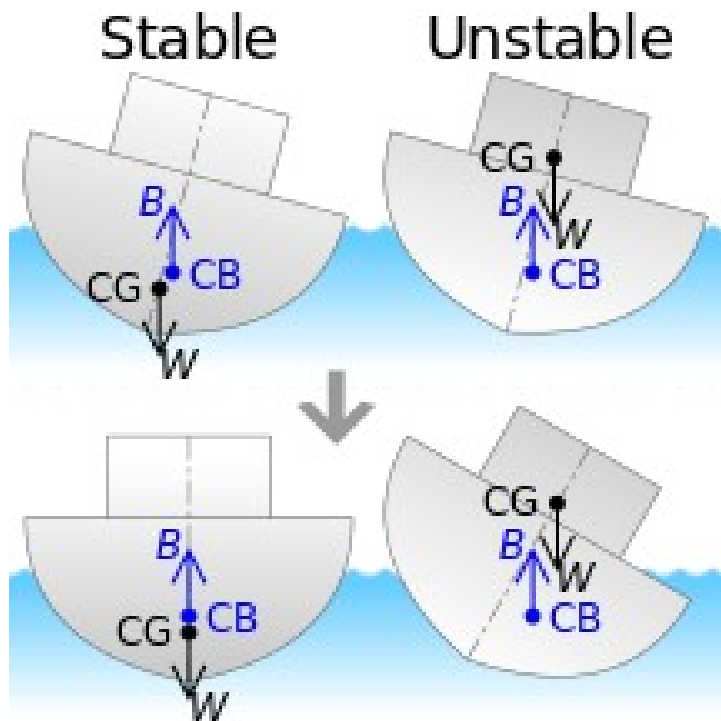


Wheel and axle

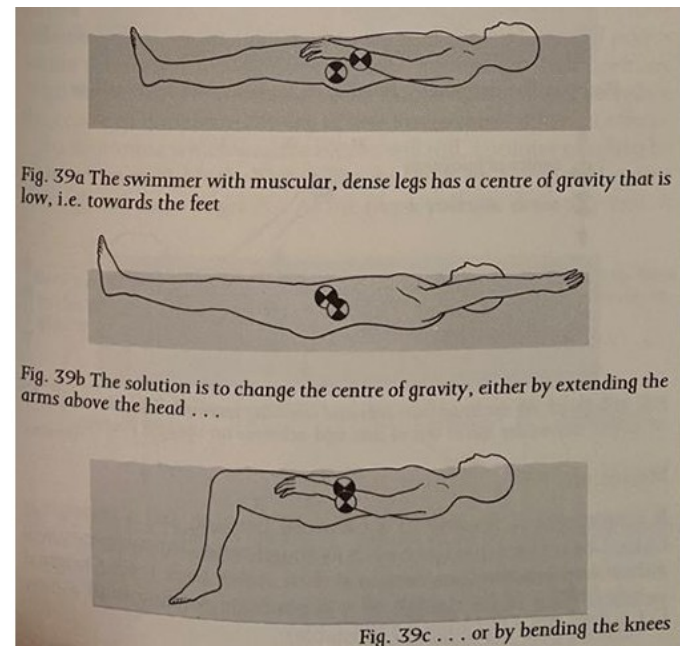


Low gear → little effort, large angle → large load , small angle

Figure 0.4 Archimedes is best known for observing that water was displaced when he stepped into his bath. He proceeded to conduct experiments where he measured the amount of water that overflowed when objects were placed into a tub full of water. He established a principle that states that an object immersed in a fluid will experience a buoyant force equal to the weight of the displaced fluid. Archimedes is also recognized for his work with simple machines like the screw, lever, and pulley. Legend has it that Archimedes said, “Give me a firm spot on which to stand and I will move the earth” (referring to the use of a lever).



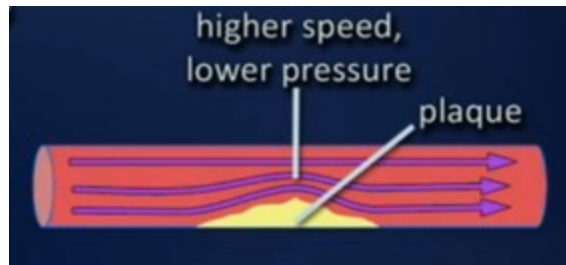
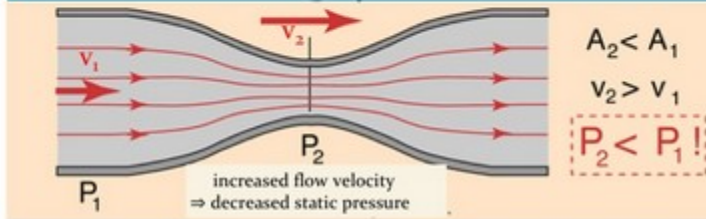
Fluid mechanics



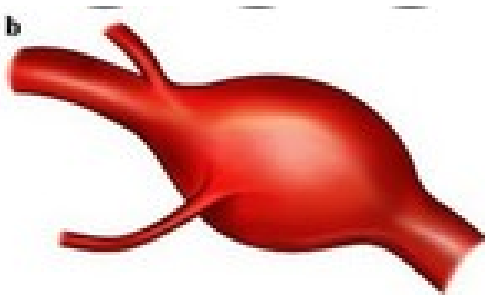
"Bernoulli's Principal"

Static Pressure 1 + Dynamic Pressure 1 = Static Pressure 2 + Dynamic Pressure 2 = Const.

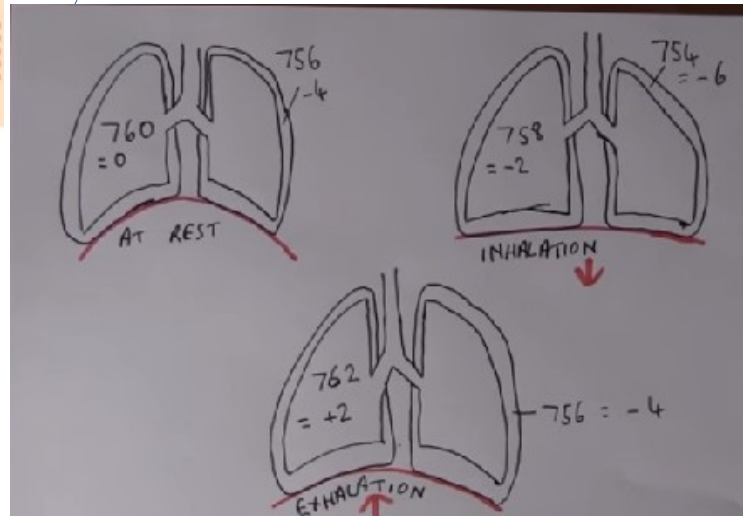
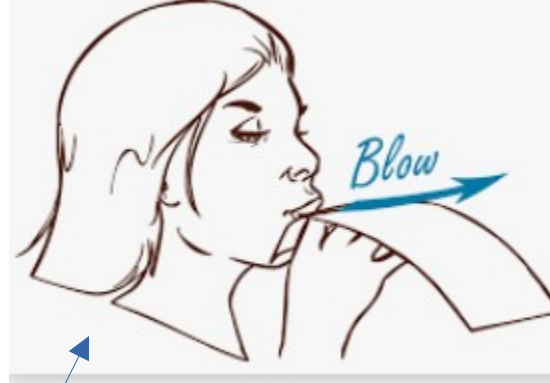
$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2 = \text{Const.}$$



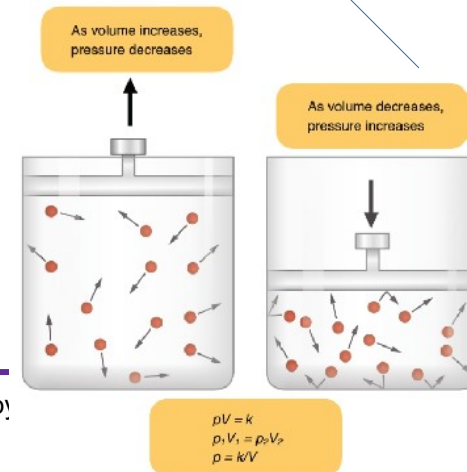
Arterial stenosis:



Fusiform Aneurysm

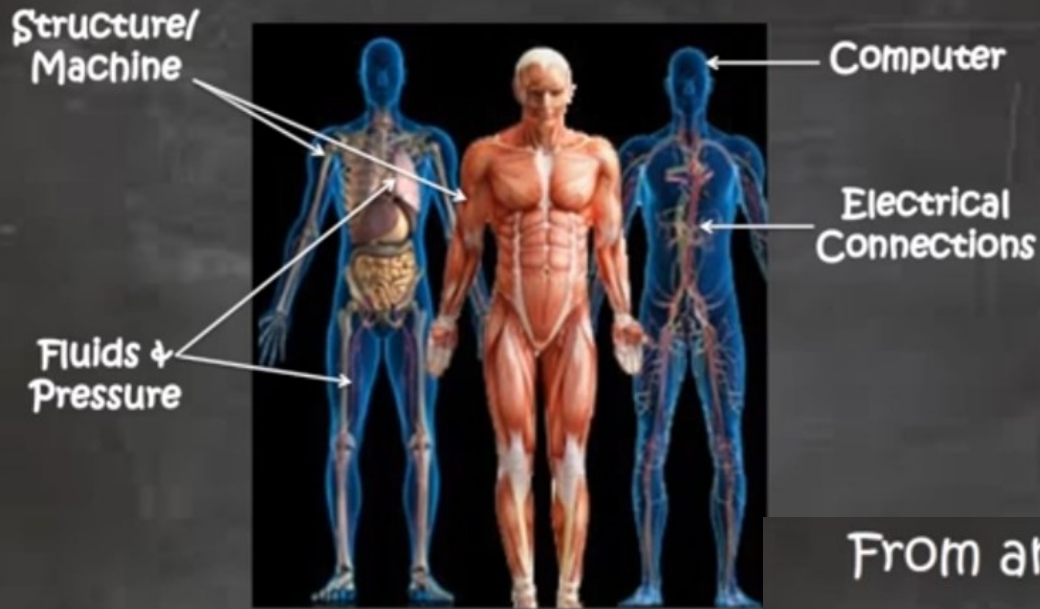


Thermodynamics Boyle's law



Copy

From an Engineering Perspective...

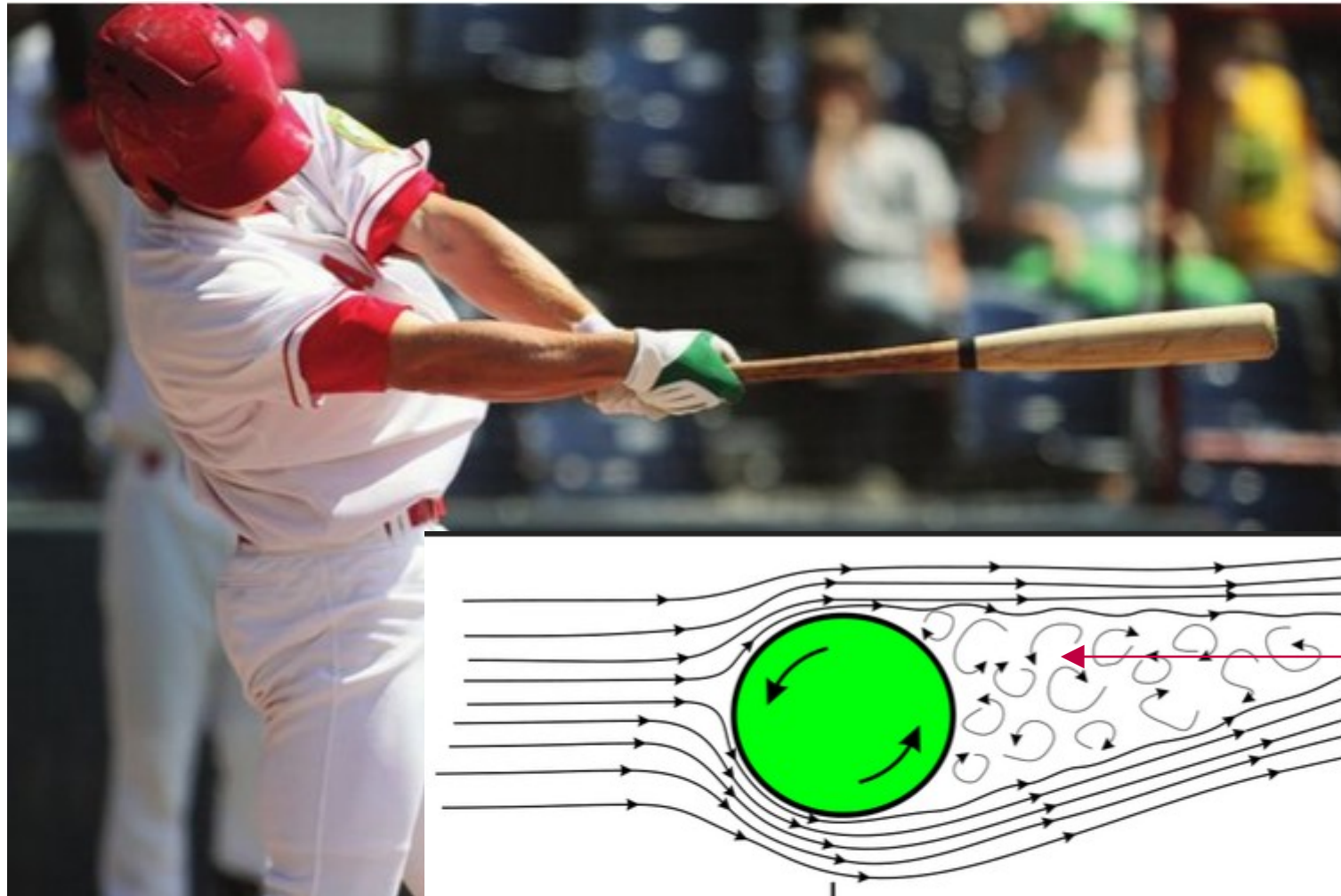


From an Engineering Perspective...

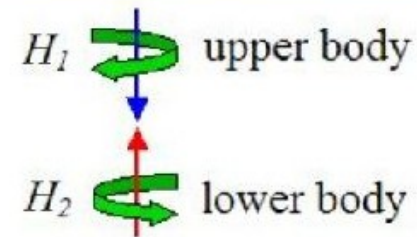
- The musculoskeletal system is a structure and machine that provides balance and motion.
- Functions of the cardiovascular system (breathing and pulse) are governed by laws of fluids and pressure.
- The brain is a computer with an electrical connection called the nervous system.

Figure 0.2 A baseball player's understanding of physics can help improve all aspects of his game, including pitching, batting, and fielding. Photo courtesy of fotolia © Paul Yates. Reprinted with permission

Video



Angular momentum, torque, conservation of momentum,



$$H_1 + H_2 = 0$$

Figure 0.14 Each ski design is created for use based on scientific knowledge. Photo courtesy of fotolia © Lulu Berlu



Friction
Air resistance



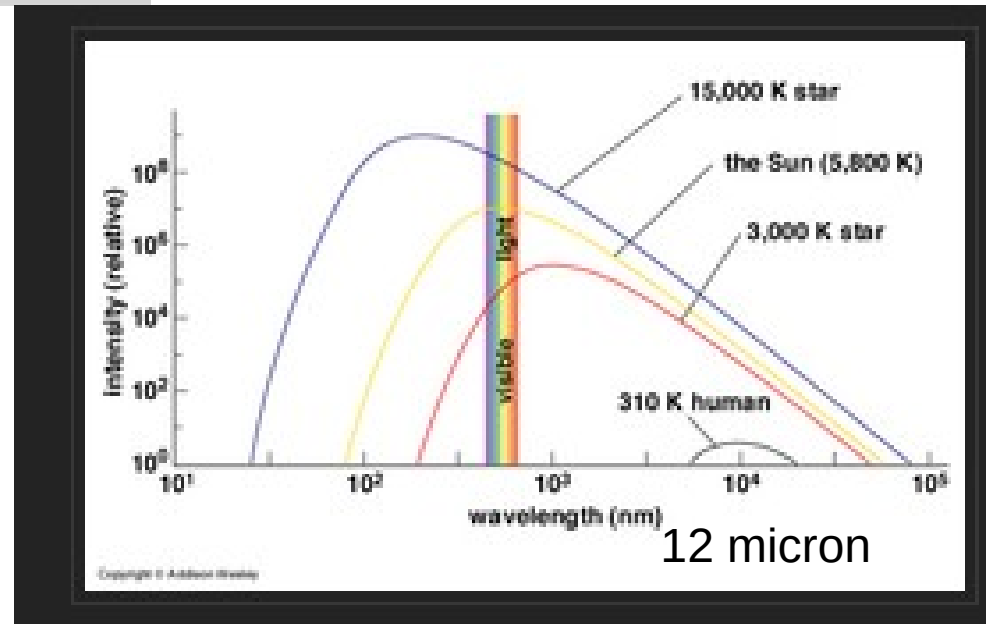


Heat

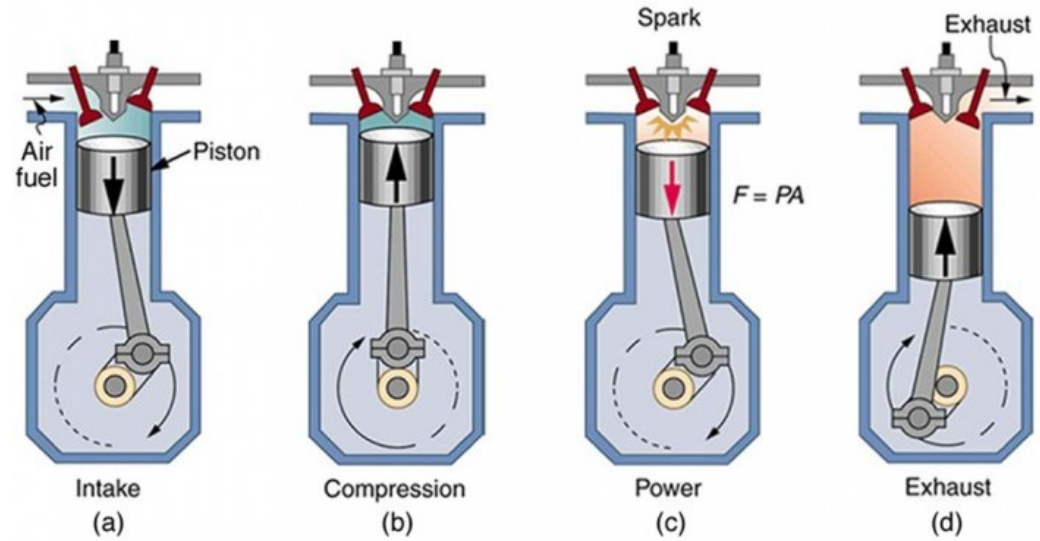
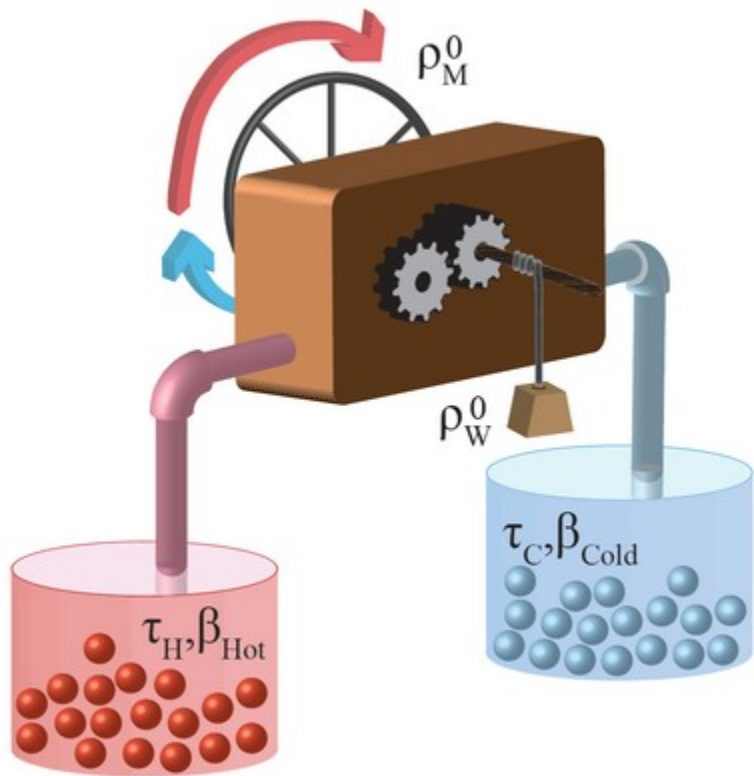
Black body

We are on average like
A 100W light bulb

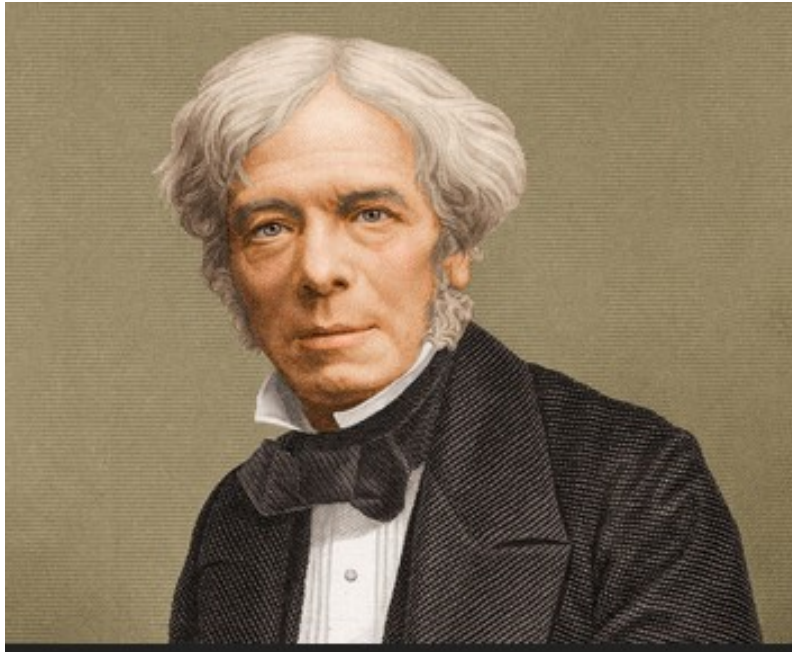
$1000 \text{ Wh} = 1000 \text{ Cal}$



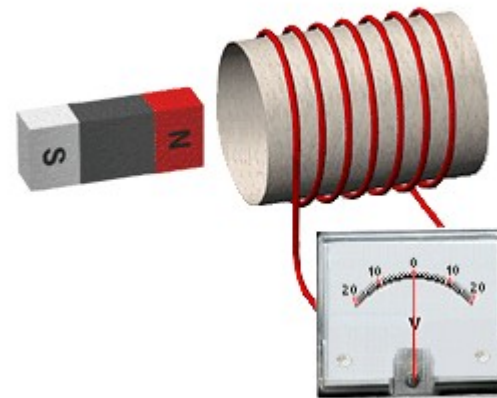
Thermodynamics → Science of the study of the flow of heat



Michael Farady



Faradays Law of Induction



Kieran Mckenzie

Figure 0.8 A magnet levitates above a superconductor, demonstrating the Meissner effect. Photo courtesy of Brookhaven National Laboratory

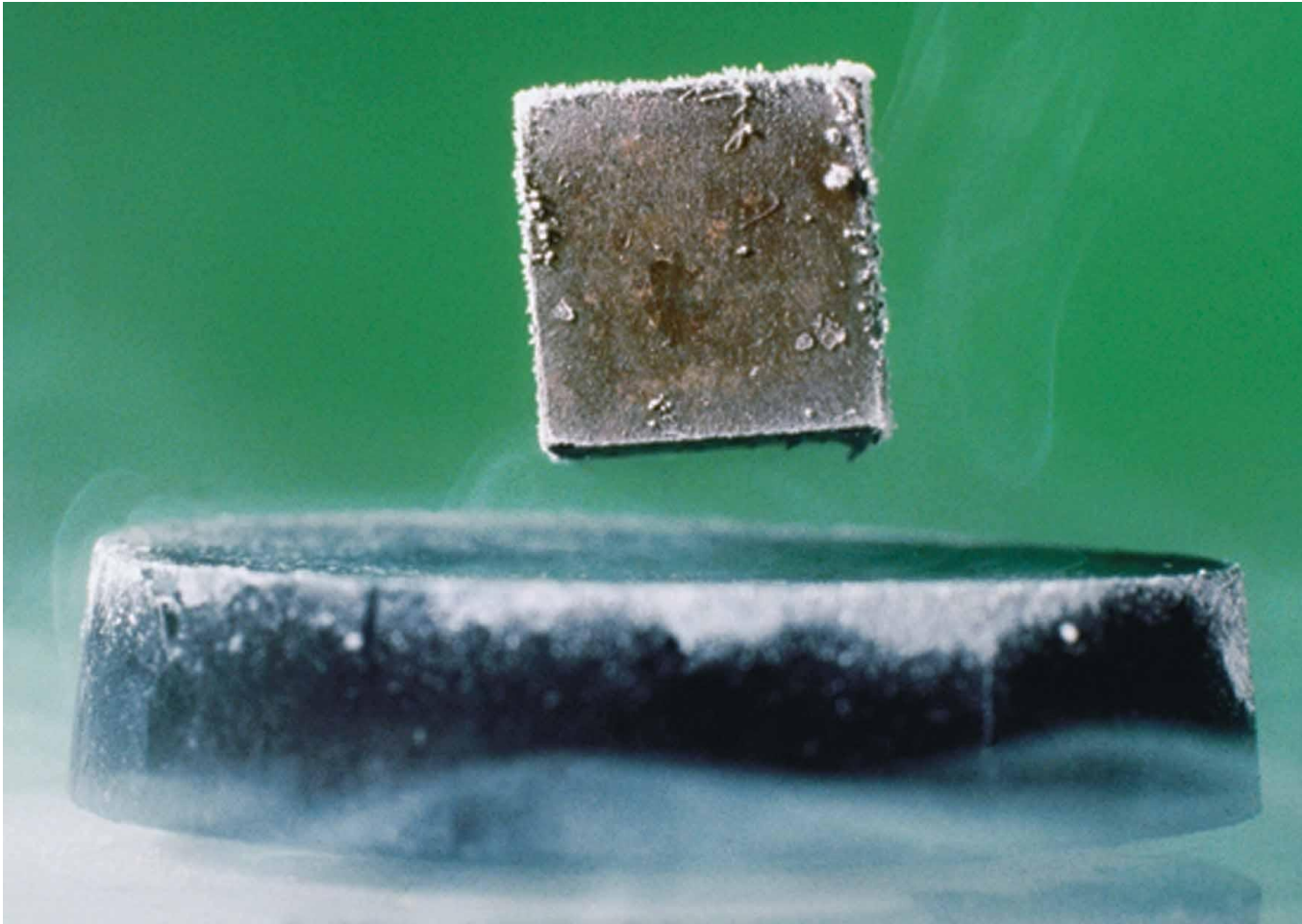


Figure 0.11 Maglev train technology has high-speed trains competing with airplane service.



Figure 0.7 The New Clark Bridge, a cable-stayed bridge in Alton, Illinois



Properties of Waves →
resonance or destructive interference



Physics of the very small → transistors
Quantum Physics

Our World
in Data

Transistor count

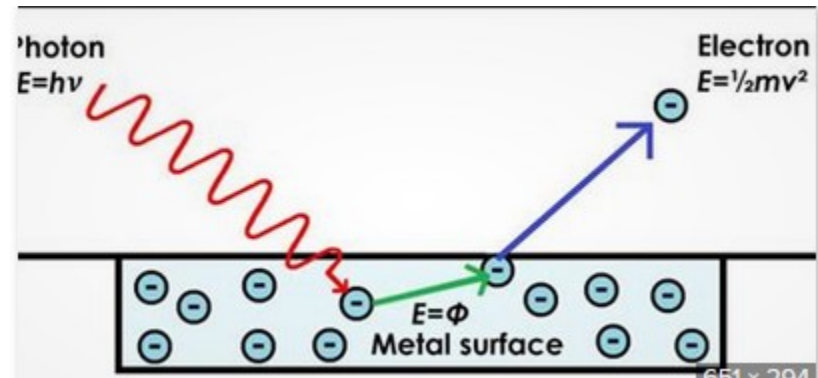


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Figure 0.1 Physics is involved in all aspects of cellular phone technology. It controls everything from the electrical circuits in the phone to the transmission of radio waves between phones.



Figure 0.3 Albert Einstein (1879–1955) is often considered one of the most influential scientists of the twentieth century. His work on relativity, as made famous through the equation $E = mc^2$, and the photoelectric effect changed the way the world viewed physics. Photo courtesy of the National Archives and Records Administration



BEWARE of pseudo Sciences !



PHOTO: ÉTIENNE KLEIN



@EtienneKlein · Follow

Photo (VRAIE cette fois...) de la galaxie de la Roue du Chariot et de ses galaxies compagnes, prise par le JWST. Située à 500 millions d'années-lumière, elle fut sans doute spirale dans son passé, mais a pris cette étrange allure à la suite d'un furieux carambolage galactique.



A photo tweeted by a famous French physicist supposedly of Proxima Centauri by the James Webb Space Telescope was actually a slice of chorizo.